A counterexample to Fisk's totally stable matrix conjecture

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Task 1. Resolve the totally stable matrix conjecture. Fisk's Conjecture 21.84 in [1], printed page 648, asserts that if $f(x,y)=\sum_{j\ge 0} f_j(x)y^j$ is a real polynomial with positive coefficients, nonvanishing when both variables lie in the open upper half-plane, then every minor of

$$T_f(x)=\begin{pmatrix} f_0 & f_1 & f_2 & f_3 & \cdots \\ 0 & f_0 & f_1 & f_2 & \cdots \\ 0 & 0 & f_0 & f_1 & \cdots \\ \vdots & \vdots & \vdots & \ddots & \ddots \end{pmatrix}$$

is stable. The same assertion appears as Question 10 of [2]. Here a univariate polynomial is *stable* if it has no zero in the open right half-plane. The conjecture is false.

An admissible input. Take

$$\begin{aligned} f(x,y) &= (10x+y+1)(x+y+10) \\ &= 10x^2+11xy+y^2+101x+11y+10. \end{aligned}$$

Every displayed coefficient is positive. If $\operatorname{Im} x, \operatorname{Im} y>0$, the imaginary parts of the two factors are $10\operatorname{Im} x+\operatorname{Im} y>0$ and $\operatorname{Im} x+\operatorname{Im} y>0$, respectively. Neither factor vanishes. Thus f belongs to the source's class $\mathbf{P}_2^{\mathrm{pos}}$. It is also nonvanishing when both variables lie in the open right half-plane, directly from the real parts of its factors.

The offending minor. Write

$$a(x)=11x+11, \qquad b(x)=10x^2+101x+10.$$

Using zero-based indices, choose rows $0,1,2,3$ and columns $1,2,3,4$ of T_f. This gives the genuine minor

$$M(x)=\begin{pmatrix} a & 1 & 0 & 0 \\ b & a & 1 & 0 \\ 0 & b & a & 1 \\ 0 & 0 & b & a \end{pmatrix}.$$

Expansion of these tridiagonal determinants gives $D_0=1$, $D_1=a$ and $D_k=aD_{k-1}-bD_{k-2}$. Consequently,

$$\begin{aligned} D(x):=\det M(x) &= a^4-3a^2b+b^2 \\ &= 11111x^4+16661x^3+17661x^2+16661x+11111. \end{aligned} \tag{1}$$

An exact certificate of failure. Define

$$G(t) = 11111t^2 + 16661t - 4561.$$

For $x \neq 0$, equation (1) gives the identity

$$D(x) = x^2 G(x + x^{-1}). \tag{2}$$

The exact evaluations

$$G(1/5) = -\frac{19609}{25} < 0, \qquad G(1/4) = \frac{4779}{16} > 0$$

show that G has a real zero $t_0 \in (1/5, 1/4)$. Set

$$z_0 = \frac{t_0 + i\sqrt{4 - t_0^2}}{2}.$$

Then $|z_0| = 1$, $z_0 + z_0^{-1} = t_0$, and

$$\frac{1}{10} < \operatorname{Re} z_0 < \frac{1}{8}.$$

Equation (2) proves $D(z_0) = 0$. Thus this minor has a zero strictly in the forbidden half-plane. Its conjugate is another such zero. The input satisfies every hypothesis, so the universal assertion is disproved. This proof uses exact integer identities and the intermediate value theorem; no approximate root calculation or boundary convention is needed.

Task 2. Determine the smallest failing minor order. Decide whether every minor of order at most three is stable for every $f \in \mathbf{P}_2^{\mathrm{pos}}$, or construct a smaller counterexample. The example above gives an upper bound of four on the minimum order.

Task 3. Classify the two-factor case. For $f(x, y) = (a_1x + b_1y + c_1)(a_2x + b_2y + c_2)$ with positive parameters, characterize when every minor of $T_f(x)$ is stable. Identify how the first failing order depends on the parameter ratios.

Task 4. Test the weaker coefficient assertion. Determine whether every minor of $T_f(x)$ has nonnegative coefficients for arbitrary $f \in \mathbf{P}_2^{\mathrm{pos}}$. The present counterexample retains coefficient positivity, so it does not decide that weaker claim.

Task 5. Find valid preservation hypotheses. Find structural conditions on polynomial matrices under which multiplication preserves stability of every minor, and use them to identify subclasses for which Fisk's original conclusion does hold.

References

[1] S. Fisk, *Polynomials, roots, and interlacing*, version 2, arXiv:math/0612833, 2008, §21.1–21.2 and Conjecture 21.84.

[2] S. Fisk, *Questions about determinants and polynomials*, arXiv:0808.1850, 2008, Definition 1 and Question 10.